Using Hamiltonians with different qubit numbers on QAOA

https://github.com/qiboteam/qibo/issues/558

ROW 57

Using Hamiltonians with different qubit numbers on QAOA #558

Closed

New issue

stavros11 opened on Mar 9, 2022

Following from #556, the following script

```
import numpy as np
from qibo import hamiltonians, models
from qibo.symbols import X, Z

h_obj = hamiltonians.SymbolicHamiltonian(Z(0)*Z(1)+Z(1)*Z(2)+Z(2)*Z(3) + Z(3)*Z(0))
h_mixer = hamiltonians.SymbolicHamiltonian(X(0)*X(1)*X(2))
qaoa = models.QAOA(h_obj, mixer=h_mixer)

initial_parameters = 0.01 * np.random.uniform(0, 1, 4)
best_energy, final_parameters, extra = qaoa.minimize(initial_parameters, method="BFGS")
```

fails with

```
ValueError: Invalid tensor shape (16,) for state of 3 qubits.
```

because the second Hamiltonian acts on three qubits while the first acts on four. I had a look at the code and I am not sure how easy it is to implement this for Hamiltonians acting on different qubit numbers.

Right now the number of qubits of a Hamiltonian is fixed:

- if the Hamiltonian is created using a matrix the number of qubits is given by the user and should be consistent with the given matrix size (which should be 2^n x 2^n),
- if the Hamiltonian is created from symbols the number of qubits is inferred from the expression. For example `X(0) * X(1)` will create two qubit Hamiltonian, `X(0) * X(1) * X(3)` four qubit, etc.

QAOA requires applying the exponential of the Hamiltonian to the state.

- if the Hamiltonian is created using matrix, this is calculated using `expm` on the full matrix and is applied to the state,
- if the Hamiltonian is created using symbols, a qibo circuit that implements the exponential via the Trotter decomposition is created and applied to the state.

In both cases, the number of qubits in the state should agree with the number of qubits in the Hamiltonian. In the first case because the matrix dimension is fixed and in the second case because the number of qubits of the Trotter circuit is the number of qubits associated with the Hamiltonian.

To fix the above QAOA issue we could do one of the following:

1. Force the user to use Hamiltonians with the same qubit numbers in the QAOA by raising an error if this condition is not satisfied. This should be simple as one can use the identity symbol implemented in Add identity symbol #557 to add qubits in the smaller Hamiltonian.
2. Automatically expand the smaller Hamiltonian using identities within the QAOA (a bit messy solution).
3. Since it is mathematically possible to multiply exp(-i H) to a state with more qubits than the support of H, we could implement this. This is more complicated to implement as we would need to change several methods in the Hamiltonian classes, not the QAOA.

@scarrazza @DiegoGM91 let me know what you think.

stavros11 added bug enhancement on Mar 9, 2022

stavros11 assigned stavros11 and DiegoGM91 on Mar 9, 2022

stavros11 changed the title ~~Using Hamiltonian with different qubit numbers on QAOA~~ Using Hamiltonians with different qubit numbers on QAOA on Mar 9, 2022

Assignees

DiegoGM91

stavros11

Labels

bug enhancement

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

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Participants

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DiegoGM91 on Mar 10, 2022 Contributor

@stavros11 thanks for looking into this.

I think solution 1 is perfectly fine, as long as it is well documented.

Solution 2 seems fine to me too from the user's perspective, but if you think it is not desirable from the developer point of view, then I think solution 1 should suffice.

DiegoGM91 on Mar 10, 2022 Contributor

Solution 3 would make sense to me only if there is a performance gain of this approach compared to defining the Hamiltonian using identity symbols.

stavros11 mentioned this on Mar 25, 2022

Raise error if QAOA Hamiltonians have different nqubits #565

scarrazza closed this as completed on Mar 30, 2022

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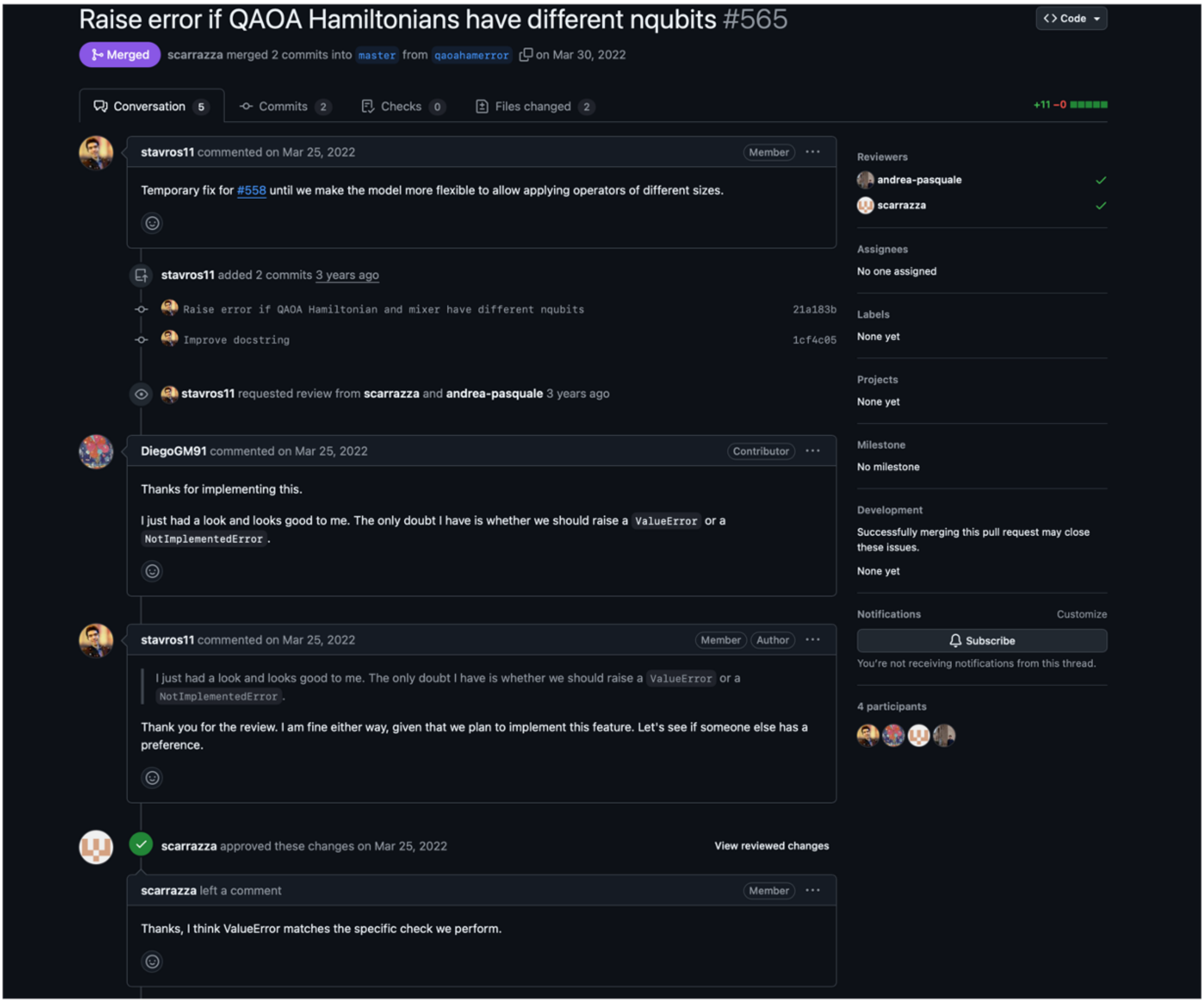

Raise error if QAOA Hamiltonians have different nqubits #565

<> Code

Merged scarrazza merged 2 commits into `master` from `qaoahamerror` on Mar 30, 2022

Conversation 5 | Commits 2 | Checks 0 | Files changed 2 | +11 −0

stavros11 commented on Mar 25, 2022 — Member

Temporary fix for #558 until we make the model more flexible to allow applying operators of different sizes.

stavros11 added 2 commits 3 years ago

- `Raise error if QAOA Hamiltonian and mixer have different nqubits` 21a183b
- `Improve docstring` 1cf4c05

stavros11 requested review from **scarrazza** and **andrea-pasquale** 3 years ago

DiegoGM91 commented on Mar 25, 2022 — Contributor

Thanks for implementing this.

I just had a look and looks good to me. The only doubt I have is whether we should raise a `ValueError` or a `NotImplementedError`.

stavros11 commented on Mar 25, 2022 — Member, Author

> I just had a look and looks good to me. The only doubt I have is whether we should raise a `ValueError` or a `NotImplementedError`.

Thank you for the review. I am fine either way, given that we plan to implement this feature. Let's see if someone else has a preference.

scarrazza approved these changes on Mar 25, 2022 — View reviewed changes

scarrazza left a comment — Member

Thanks, I think ValueError matches the specific check we perform.

Reviewers
andrea-pasquale ✓
scarrazza ✓

Assignees
No one assigned

Labels
None yet

Projects
None yet

Milestone
No milestone

Development
Successfully merging this pull request may close these issues.
None yet

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4 participants